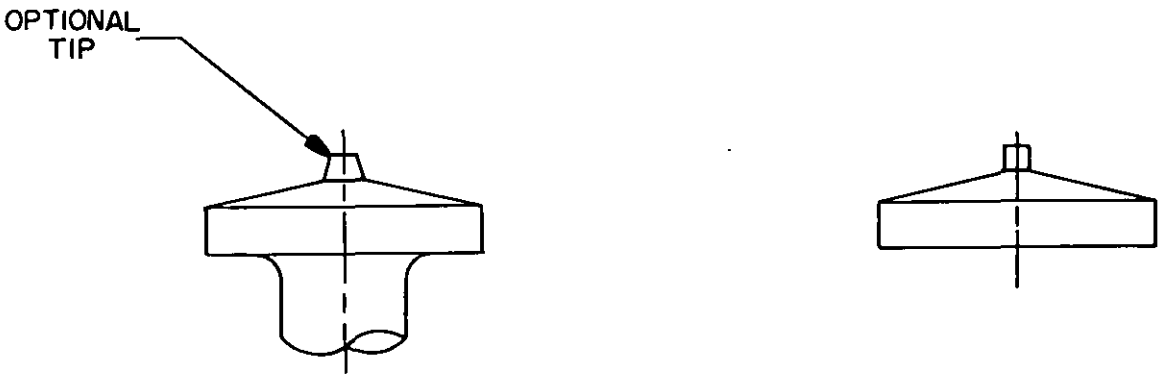
INCH-POUND
MIL-S-24149/5C(SH)
27 March 1989
SUPERSEDING
MIL-S-24149/5B
1 June 1981

MILITARY SPECIFICATION SHEET

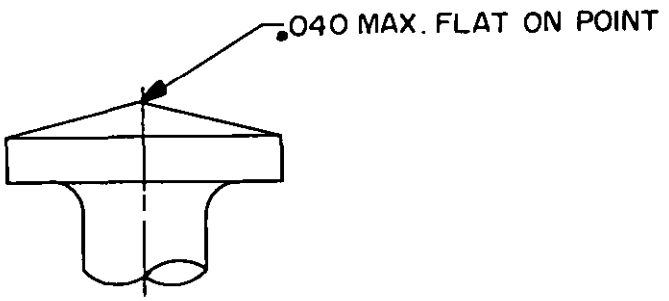
# STUDS, WELDING, TYPE VII, CLASS 1,2,3, ALUMINUM ALLOY, FOR STORED ENERGY (CAPACITOR DISCHARGE) ARC WELDING

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-S-24149(SH).




ALUMINUM ALLOY STUDS, TYPE VII, STYLE A, TIPPED WELD-END.




ALUMINUM ALLOY STUDS, TYPE VII, STYLE B, POINTED WELD-END.

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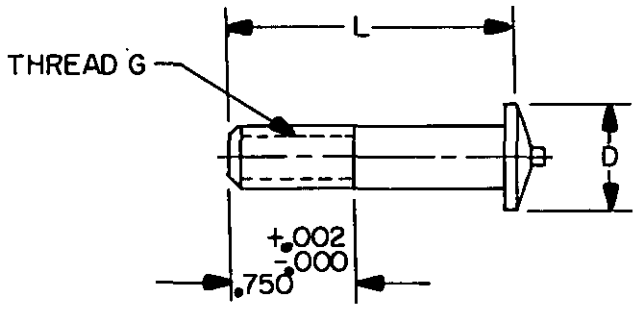
NOTE:
1. The choice of weld-end style shall be based on the type of stored energy welding equipment to be used.

FIGURE 1. Optional weld-end styles.

AMSC N/A
FSC 5307
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SH 11917

FIGURE 2. Type VII, class 1, stud with threaded section.

TABLE I. Configuration part numbers.

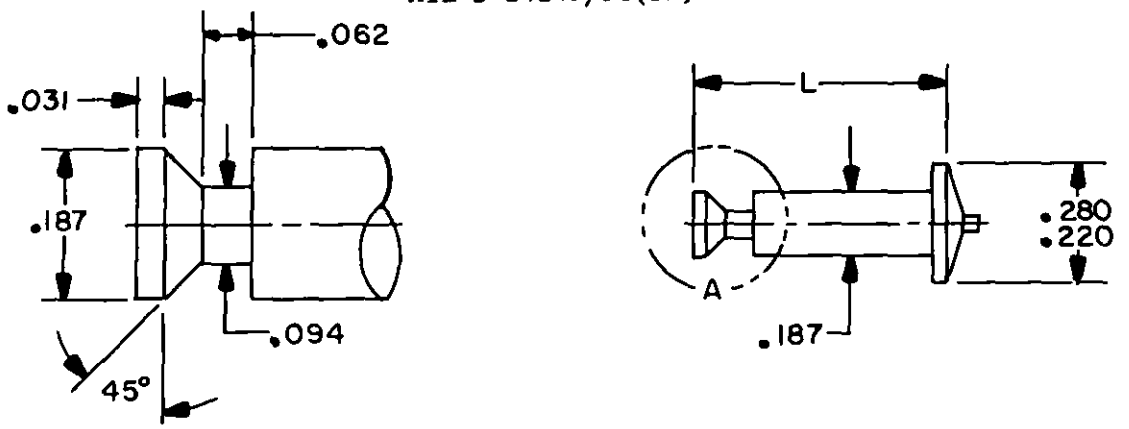
| G | D | L ±0.031 | M24149/5 dash no. |
|---|---|---|---|
| 0.1900-32UNF-2A | 0.280/.220 | 1.250 | -1 |
| | | 1.375 | -2 |
| | | 1.500 | -3 |
| | | 1.625 | -4 |
| | | 1.750 | -5 |
| | | 1.875 | -6 |
| | | 2.000 | -7 |
| | | 2.250 | -8 |
| | | 2.500 | -9 |
| | | 2.750 | -10 |
| | | 3.000 | -11 |
| | | 3.250 | -12 |
| | | 3.500 | -13 |
| 0.2500-20UNC-2A | 0.340/.280 | 1.250 | -14 |
| | | 1.375 | -15 |
| | | 1.500 | -16 |
| | | 1.625 | -17 |
| | | 1.750 | -18 |
| | | 1.875 | -19 |
| | | 2.000 | -20 |
| | | 2.250 | -21 |
| | | 2.500 | -22 |
| | | 2.750 | -23 |
| | | 3.000 | -24 |
| | | 3.250 | -25 |
| | | 3.500 | -26 |

NOTE:

1. Dimensions are in inches. Unless otherwise specified, tolerance: decimals plus or minus 0.015.



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VIEW A

FIGURE 3. Type VII, class 2, stud with notched end.

TABLE II. Length part numbers.

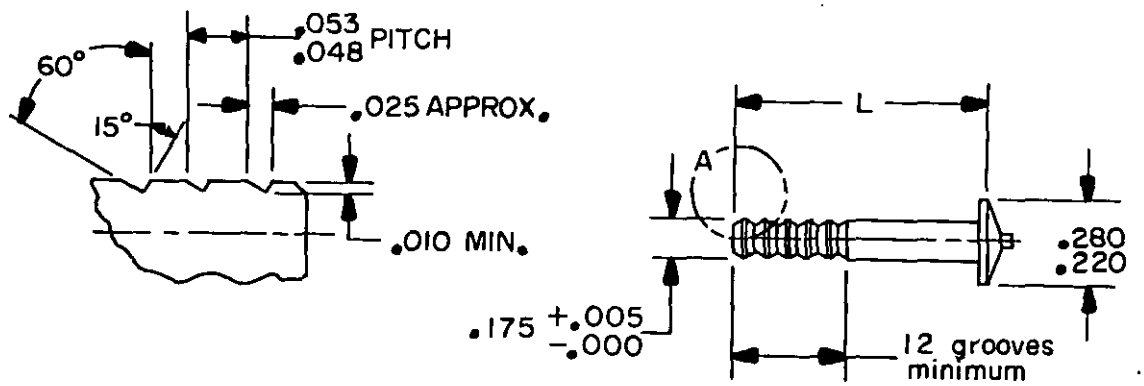
| L ±0.031 | M24149/5 dash no. |
|---|---|
| .625 | -101 |
| .750 | -102 |
| .875 | -103 |
| 1.000 | -104 |
| 1.125 | -105 |
| 1.250 | -106 |
| 1.375 | -107 |
| 1.500 | -108 |
| 1.625 | -109 |
| 1.750 | -110 |
| 1.875 | -111 |
| 2.000 | -112 |
| 2.250 | -113 |
| 2.500 | -114 |
| 2.750 | -115 |
| 3.000 | -116 |

NOTE:

1. Dimensions are in inches. Unless otherwise specified, tolerances: decimals plus or minus 0.015, angles plus or minus 2 degrees.



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SH 11919 VIEW A

FIGURE 4. Type VII, class 3, stud with grooves (annular rings).

TABLE III. Length part numbers.

| L ±0.031 | M24149/5 dash no. |
|---|---|
| .625 | -201 |
| .750 | -202 |
| .875 | -203 |
| 1.000 | -204 |
| 1.125 | -205 |
| 1.250 | -206 |
| 1.375 | -207 |
| 1.500 | -208 |
| 1.625 | -209 |
| 1.750 | -210 |
| 1.875 | -211 |
| 2.000 | -212 |
| 2.250 | -213 |
| 2.500 | -214 |
| 2.750 | -215 |
| 3.000 | -216 |

NOTE:

1. Dimensions are in inches. Unless otherwise specified, tolerances: decimals plus or minus 0.015, angles plus or minus 1 degree.



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REQUIREMENTS:

1. Material - Aluminum alloy.
2. Tensile strength - 35,000 lb/in$^2$, minimum
3. Chemical composition - Studs shall be of the chemical compositions indicated in table IV.

TABLE IV. Chemical composition (single values are maximum).

| Alloy no. | Copper | Mag-nesium | Man-ganese | Sil-icon | Iron | Zinc | Ti-tanium | Chro-mium | Others | | Alum-inum |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Each | Total | |
| 5083 | 0.10 | 4.0/4.9 | 0.30/1.0 | 0.40 | 0.40 | 0.25 | 0.15 | 0.05/.25 | 0.05 | 0.15 | Remainder |
| 5086 | .10 | 3.5/4.5 | .20/.70 | .40 | .50 | .25 | .15 | .05/.25 | .05 | .15 | Remainder |
| 5356 | .10 | 4.5/5.5 | .05/.20 | .25 | .40 | .10 | .06/.20 | .05/.20 | .05 | .15 | Remainder |
| 5456 | .10 | 4.7/5.5 | .50/1.0 | .25 | .40 | .25 | .20 | .05/.20 | .05 | .15 | Remainder |

4. Part number - The part number consists of the prefix letter "M" and the specification sheet number plus the applicable dash number.

Example:
M24149/5-7 - Type VII, class 1 welding stud 0.1900-32 UNF-2A thread, 2.000 inch length.

NOTE:

1. Changes from previous issue - Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5307-N028)



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